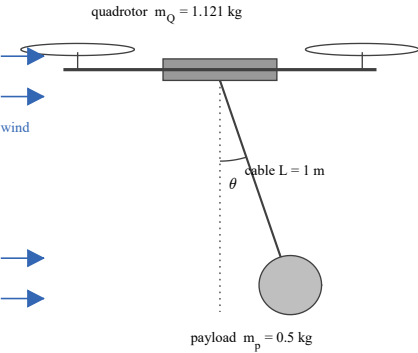


(a) plant P2 (side view)



wind: measured NREL M5 series (20 Hz), acting on airframe and payload

quadratic drag on the velocity relative to the air:

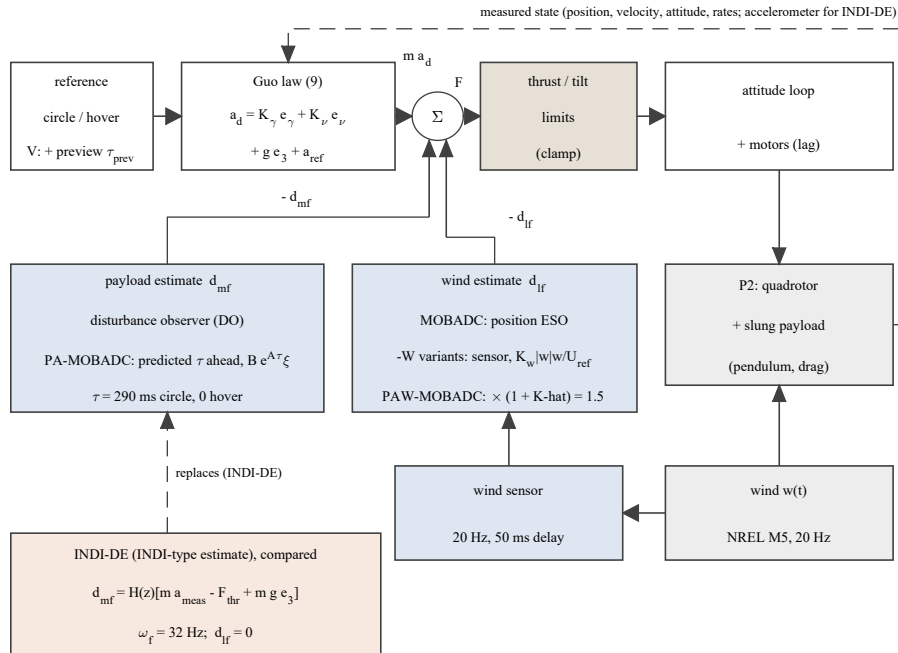
$$F = K_w |v| v / U_{\text{ref}}, \quad K_w = 0.2 \text{ N s/m}, \quad U_{\text{ref}} = 5 \text{ m/s}$$

$$\text{payload drag area } (C_{D^A})_L = K (C_{D^A})_Q, \quad K = 0.5$$

wind sensor 20 Hz, delay 50 ms; motors: first-order lag 17 ms

limits: $f_i \leq 7.668 \text{ N}$ per rotor, total $\leq 27.6 \text{ N}$, tilt $\leq 30^\circ$

(b) controller and the compared estimates



controllers (circle; hover uses its exact-frequency DO table):

MOBADC: wind ESO, DO harmonics $\{1\}$ MOBADC-W: wind sensor, DO $\{0, 1\}$ + preview: $\tau_{\text{prev}} = 180 \text{ ms}$

PA-MOBADC: MOBADC-W + payload estimate predicted τ ahead (C1)

PAW-MOBADC (proposed): PA-MOBADC + static $(1 + K\text{-hat})$ payload-wind feed-forward (C2) INDI-DE: wind channel 0